ETIOLOGY AND PATHOGENESIS

TSC is caused by mutations in a tumor suppressor gene, either **TSC1** or **TSC2**. Mutations in **TSC2** are observed in about three-fourths of patients, and even more commonly in de novo cases. Patients with mutations in **TSC2** tend to exhibit a more severe phenotype.

TSC1 maps to chromosome band **9q34**. The 8.6-kb full-length transcript encodes a protein called **hamartin** (TSC1). **TSC2** maps to chromosome band **16p13.3**. The 5.5-kb transcript encodes a protein called **tuberin** (TSC2). Immediately adjacent to **TSC2** on chromosome 16 is **PKD1**, the gene mutated in polycystic kidney disease. Some patients with TSC have severe, early-onset renal cystic disease, and most of these patients have a contiguous deletion of **TSC2** and **PKD1**.

Typical of mutations in tumor suppressor genes, the mutations observed in patients with TSC are inactivating mutations located anywhere along the sequence of **TSC1** or **TSC2**. Consistent with Knudson's two-hit hypothesis, most TSC tumors show a second somatic mutation that inactivates the wild-type allele. **TSC1** and **TSC2** form a complex that inhibits signaling through the mammalian target of rapamycin (**mTOR**) pathway. Loss of TSC1/TSC2 function leads to increased mTOR signaling and increased cell growth. **Rapamycin**, a drug that inhibits mTOR, may be useful for the treatment of internal tumors in TSC.

CLINICAL FINDINGS

Clinical Criteria

The diagnosis of TSC is based on clinical criteria categorized as **major** or **minor** features. Cutaneous and oral lesions comprise 4 of 11 major features and 3 of 9 minor features. Major features are considered highly specific for TSC, whereas minor features are less specific or less substantiated. No single feature is present in all patients, and no feature is entirely specific for TSC. Genetic testing can be offered and may be important in certain clinical situations (see Molecular Diagnosis).

History

The medical and family history should include questions about seizures, intellectual disability, behavioral disorders, visual problems, and tumors of the brain, heart, kidneys, lungs, and skin. Determining the age of onset and evolution of skin lesions can help differentiate TSC-associated lesions from other similar dermatologic conditions. Many TSC skin lesions are not present at birth but appear at ages typical for each lesion type. For example, infants often present with multiple hypomelanotic macules but not angiofibromas or ungual fibromas. In contrast, adults may have multiple angiofibromas and ungual fibromas, while hypomelanotic macules may have faded or become less apparent.

Cutaneous and Oral Lesions

HYPOMELANOTIC MACULES

Hypomelanotic macules (Fig.